sky130_rhbd_tt_1P8_25C.ccs Library

Cell Groups
AND2X1
AND3X1
AO3X1
AOA4X1
AOAI4X1
AOI3X1
BUFX1
DFFQNX1
DFFQX1
DFFRNQNX1
DFFRNQX1
DFFRNX1
DFFSNQNX1
DFFSNQX1
DFFSNRNQX1
DFFSNRNX1
DFFSNX1
DFFX1
DLATCHN
DLATCH
FA
HA
INVX1

AND2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
AND2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
AND2X1	0.01067	0.01080	6.00091

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND2X1	0.00000	7.37298	13.28970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (RR)	0.07188	0.64093	7.36437
	B->Y (RR)	0.06518	0.64365	7.38134

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (FF)	0.06112	0.55496	6.28215
	B->Y (FF)	0.05884	0.55588	6.09628

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	8.25140	8.26571	8.56157
	B	0.00000	0.00000	0.00000
	B	8.25211	8.26733	8.57161

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	2.39215	2.41248	2.76296
	B	0.00000	0.00000	0.00000
	B	2.38432	2.39956	2.67165

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.55824	2.55773	2.55761

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.15043	3.14984	3.14991

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.56108	2.56058	2.56098

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.15110	3.15112	3.15158

AND3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	0
1	0	x	0
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AND3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AND3X1	0.01068	0.01048	0.01071	6.07005

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND3X1	0.00000	6.97639	20.76320

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (RR)	0.10035	0.67974	7.62950
	B->Y (RR)	0.09359	0.68724	7.65694
	C->Y (RR)	0.08455	0.69301	7.72042

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (FF)	0.06829	0.57458	6.35470
	B->Y (FF)	0.06636	0.56932	6.08847
	C->Y (FF)	0.06272	0.57015	6.07302

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	13.03900	13.04900	13.32060
	B	0.00000	0.00000	0.00000
	B	13.03960	13.04990	13.31820
	C	0.00000	0.00000	0.00000
	C	13.04020	13.05170	13.32360

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	1.66299	1.68266	2.02986
	B	0.00000	0.00000	0.00000
	B	1.65461	1.66992	1.95104
	C	0.00000	0.00000	0.00000
	C	1.64648	1.65885	1.88620

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.22206	2.22144	2.22118
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.22421	2.22293	2.22358
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22058	2.22032	2.22024

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.78057	2.77972	2.77988
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.78183	2.78130	2.78120
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.77888	2.77861	2.77896

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.22206	2.22181	2.22180
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22739	2.22680	2.22654
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.22067	2.21977	2.22057

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.78016	2.77932	2.77941
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.78215	2.78240	2.78225
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.78026	2.78010	2.78056

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.22512	2.22434	2.22459
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.22815	2.22640	2.22769

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.78094	2.78144	2.78109
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.78043	2.78024	2.78034

AO3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AO3X1	76.51360

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AO3X1	0.01029	0.01056	0.01056	5.69722

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AO3X1	0.00000	5.28280	19.23130

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (FR)	0.10430	0.74570	7.65060
	B->Y (FR)	0.10042	0.72120	7.30907
	C->Y (RR)	0.05405	0.61500	6.79931

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (RF)	0.14529	0.59355	5.04974
	B->Y (RF)	0.13735	0.58360	4.95047
	C->Y (FF)	0.08143	0.59935	6.18054

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	1.49678	1.51402	1.77617
	B	0.00000	0.00000	0.00000
	B	1.48717	1.50164	1.72737
	C	0.00000	0.00000	0.00000
	C	11.37980	11.39030	11.61540

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	10.20510	10.21500	10.40070
	B	0.00000	0.00000	0.00000
	B	10.20890	10.21830	10.41830
	C	0.00000	0.00000	0.00000
	C	11.39440	11.40750	11.63140

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	11.81730	11.82580	12.06010
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.03476	0.03473	0.03471

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06924	0.08430	0.34459
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.05832	0.05791	0.05787

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	11.81700	11.82740	12.06570
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.03739	0.03735	0.03734

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06043	0.07263	0.29916
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.05813	0.05834	0.05828

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * !B * Y) + (!A * Y)	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	0.02397	0.02397	0.02393

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	**(A * !B * Y) + (!A * Y)**	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	1.47393	1.47384	1.47397

AOA4X1

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Truth Table

INPUT				OUTPUT
A	B	C	D	Y
0	x	x	x	0
1	0	x	x	0
1	1	x	0	0
1	1	0	1	1
1	1	1	1	0

Footprint

Cell Name	Area
AOA4X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	Y
AOA4X1	0.01028	0.01052	0.01048	0.01055	5.51490

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOA4X1	0.00000	10.36610	25.24350

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (RR)	0.20032	0.77139	6.77454
	B->Y (RR)	0.19252	0.77618	6.87993
	C->Y (FR)	0.13541	0.77622	7.69466
	D->Y (RR)	0.06803	0.64792	7.01083

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (FF)	0.14307	0.62333	5.39139
	B->Y (FF)	0.13943	0.61336	5.27506
	C->Y (RF)	0.09256	0.50239	4.34219
	D->Y (FF)	0.05996	0.54917	5.68179

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	14.30610	14.31290	14.53890
	B	0.00000	0.00000	0.00000
	B	14.30960	14.31560	14.58980
	C	0.00000	0.00000	0.00000
	C	15.16130	15.17340	15.37410
	D	0.00000	0.00000	0.00000
	D	15.24190	15.25240	15.50100

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	4.88106	4.89642	5.19597
	B	0.00000	0.00000	0.00000
	B	4.87281	4.88610	5.14851
	C	0.00000	0.00000	0.00000
	C	14.49970	14.50730	14.70950
	D	0.00000	0.00000	0.00000
	D	9.30818	9.32170	9.55464

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	15.38080	15.38860	15.61830
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	8.11289	8.11801	8.30200
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.75287	2.75281	2.75269

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	2.51160	2.52614	2.77991
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	5.02269	5.03679	5.27783
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.06289	3.06247	3.06239

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	15.38070	15.38970	15.62310
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	8.11960	8.12353	8.31390
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.75415	2.75377	2.75437

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	2.50142	2.51544	2.74504
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	5.01495	5.02679	5.24480
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.06297	3.06301	3.06311

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * !D * !Y)	0.00000	0.00000	0.00000
	(A * B * !D * !Y)	15.21270	15.22080	15.42700
	(A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y) + (!A * !Y)	1.97982	1.97950	1.97966

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * !D * !Y)	0.00000	0.00000	0.00000
	(A * B * !D * !Y)	9.30712	9.31718	9.51923
	(A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y) + (!A * !Y)	5.01023	5.00987	5.01020

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	14.46450	14.46370	14.46420

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	0.00000	0.00000	0.00000
	(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)	15.10390	15.10400	15.10390

AOAI4X1

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Truth Table

INPUT				OUTPUT
A	B	C	D	YN
0	x	x	x	1
1	0	x	x	1
1	1	x	0	1
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
AOAI4X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	YN
AOAI4X1	0.01026	0.01052	0.01053	0.01050	5.27895

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOAI4X1	0.00000	5.05158	26.53990

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (FR)	0.11428	0.73267	7.14420
	B->YN (FR)	0.10969	0.70820	6.83771
	C->YN (RR)	0.06291	0.60446	6.32940
	D->YN (FR)	0.03112	0.79527	10.19280

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (RF)	0.16341	0.80049	7.49367
	B->YN (RF)	0.15515	0.78234	7.30417
	C->YN (FF)	0.09877	0.80110	8.42497
	D->YN (RF)	0.03300	0.80421	10.27340

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	1.33456	1.35001	1.60398
	B	0.00000	0.00000	0.00000
	B	1.32636	1.33980	1.56504
	C	0.00000	0.00000	0.00000
	C	9.89431	9.90210	10.12100
	D	0.00000	0.00000	0.00000
	D	5.16899	5.17130	5.22506

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	15.44050	15.44790	15.64040
	B	0.00000	0.00000	0.00000
	B	15.44210	15.44950	15.65430
	C	0.00000	0.00000	0.00000
	C	16.30610	16.31970	16.53030
	D	0.00000	0.00000	0.00000
	D	16.33050	16.33170	16.35100

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	10.88330	10.89090	11.11590
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	4.22329	4.22638	4.40557
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.03321	0.03320	0.03318

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	0.07060	0.08619	0.35181
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	1.41045	1.42533	1.66445
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.05454	0.05410	0.05409

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	10.88310	10.89270	11.12360
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	4.22827	4.23275	4.41882
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.03525	0.03520	0.03524

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	0.06239	0.07494	0.30869
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	1.40242	1.41446	1.63084
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.05446	0.05466	0.05460

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	10.82240	10.83130	11.03900
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	0.02648	0.02641	0.02644

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	5.42359	5.43311	5.64020
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	1.37278	1.37265	1.37285

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.13000	10.12850	10.13020

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.75620	10.75630	10.75620

AOI3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	x	x	0
1	0	x	0
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
AOI3X1	59.28640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
AOI3X1	0.01037	0.01062	0.01049	2.87215

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOI3X1	0.00000	4.61712	20.74590

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (RR)	0.10723	0.84147	7.14517
	B->YN (RR)	0.10116	0.84971	7.22976
	C->YN (FR)	0.04575	0.92167	9.98232

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (FF)	0.07113	0.45291	3.79060
	B->YN (FF)	0.06806	0.44256	3.53513
	C->YN (RF)	0.02411	0.50686	5.57924

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	5.57868	5.58926	5.82295
	B	0.00000	0.00000	0.00000
	B	5.58140	5.59202	5.83221
	C	0.00000	0.00000	0.00000
	C	6.78399	6.78722	6.84462

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	1.63354	1.65158	1.96914
	B	0.00000	0.00000	0.00000
	B	1.62526	1.63911	1.89198
	C	0.00000	0.00000	0.00000
	C	12.73880	12.74030	12.77190

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	13.00790	13.01750	13.27220
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.03500	0.03498	0.03496

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.06343	0.07628	0.32219
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.06079	0.06038	0.06034

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	13.00740	13.01810	13.27270
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.03698	0.03692	0.03691

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.05510	0.06594	0.27512
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.06089	0.06111	0.06105

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	0.01823	0.01821	0.01818

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	1.60726	1.60659	1.60685

BUFX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT
A
x

Footprint

Cell Name	Area
BUFX1	42.05920

Pin Capacitance Information

Cell Name	Pin Cap(pf)
	A
BUFX1	0.01028

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
BUFX1	0.00000	6.07079	6.07079

Passive Power Information

Passive power(pJ) for A rising :

Cell Name	Power(pJ)		
	first	mid	last
BUFX1	0.00000	0.00000	0.00000
	3.51730	3.53633	3.83858

Passive power(pJ) for A falling :

Cell Name	Power(pJ)		
	first	mid	last
BUFX1	0.00000	0.00000	0.00000
	3.53654	3.55940	3.86166

DFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
DFFQNX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
DFFQNX1	0.01055	0.02311	5.42591

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQNX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	**CLK->QN (RR)**	0.15816	0.77354	7.17577

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	**CLK->QN (RF)**	0.17629	0.87290	7.85531

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	0.05038	0.08073	0.76925
	setup	CLK (R)	0.10120	0.15817	0.75520

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	-0.03413	-0.08682	-0.73978
	setup	CLK (R)	0.06093	0.13189	1.56638

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.10852	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.18277	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.09615	0.66040	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25660	26.26570	26.61580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13080	21.13500	21.34910

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	24.93610	24.93560	24.93790
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	21.09440	21.10050	21.31620
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29550	11.30240	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	25.56960	25.56910	25.56920
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	10.10440	10.11320	10.34910
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30960	11.31810	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	21.09570	21.09810	21.30850
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	26.21560	26.21820	26.41830

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	10.32690	10.33310	10.54020
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	10.11630	10.12250	10.35220
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	9.90683	9.91143	10.11400
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	11.32150	11.33020	11.55020

DFFQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	R	0
1	R	1
x	x	IQ

Footprint

Cell Name	Area
DFFQX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
DFFQX1	0.01057	0.02311	5.42070

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	**CLK->Q (RR)**	0.13097	0.77241	7.17777

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	**CLK->Q (RF)**	0.20869	0.88075	7.84306

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	0.04637	0.07673	0.71013
	setup	CLK (R)	0.10638	0.17390	0.78451

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	-0.03287	-0.08682	-0.73777
	setup	CLK (R)	0.05791	0.12541	1.53202

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.11595	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.13575	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.19019	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.09367	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13030	21.14170	21.49430

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25770	26.26390	26.46880

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	21.09430	21.10040	21.31600
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	24.93600	24.93560	24.93790
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29350	11.29980	11.50830

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	10.10440	10.11320	10.34910
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	25.56960	25.56910	25.56920
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30760	11.31630	11.53250

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	21.09670	21.09930	21.30850
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	26.21560	26.21750	26.41790

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	10.11820	10.12550	10.35450
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	10.32680	10.33320	10.54000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	11.31880	11.32890	11.54880
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.90880	9.91450	10.11610

DFFRNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	1
x	1	x	IQN

Footprint

Cell Name	Area
DFFRNQNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	QN
DFFRNQNX1	0.01013	0.03124	0.02338	3.68547

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RR)	0.17024	0.68437	5.12498
	RN->QN (FR)	0.04918	0.73691	8.48475

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RF)	0.24719	1.01428	7.91075

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	0.03757	0.06023	0.83387
	setup	CLK (R)	0.13294	0.17426	0.60095

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	-0.02287	-0.08073	-0.69591
	setup	CLK (R)	0.06937	0.14326	1.88613

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	0.03757	0.06023	0.83387
	setup	CLK (R)	RN	0.13294	0.17426	0.60095

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	-0.02287	-0.08073	-0.69591
	setup	CLK (R)	RN	0.06937	0.14326	1.88613

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	recovery	CLK (R)	0.12323	0.19579	7.03320
	removal	CLK (R)	0.03486	0.04036	0.01790

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	recovery	CLK (R)	D	0.12323	0.19579	7.03320
	removal	CLK (R)	D	0.03486	0.04036	0.01790

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	RN ()	(CLK * D)	0.12090	0.66040	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.09120	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.07140	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.07140	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.16544	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.15307	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.21741	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10605	0.66040	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78410	23.79110	24.08500
	RN	24.23260	24.24300	24.51410

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51680	34.51840	34.70790

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	22.75700	22.75620	22.75800
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	34.32610	34.32950	34.52350
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	22.74020	22.74330	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	18.84320	18.84670	19.03820
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	9.85933	9.85943	9.85927

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	23.36840	23.36980	23.37000
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	16.53660	16.54500	16.76100
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	23.35970	23.36090	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	13.88380	13.89270	14.10360
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	10.43050	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	22.01840	22.01440	22.01630
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	22.01000	22.00810	22.00910
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	13.82010	13.82370	14.02080
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	9.22419	9.22482	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	23.85300	23.85400	23.85450
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	23.85370	23.85490	23.85550
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	10.54240	10.55040	10.74640
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	10.91460	10.91270	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	34.32870	34.32740	34.51390
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	23.89430	23.89530	24.08350
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	23.90720	23.90760	24.10180
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	23.89200	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * QN)	0.00000	0.00000	0.00000
	(D * RN * QN)	13.80890	13.81460	13.99940
	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	16.55520	16.56010	16.77150
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	9.21176	9.21577	9.40994
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	9.21746	9.22343	9.42132
	(!D * RN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !QN)	13.90110	13.90990	14.13170
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	9.21279	9.22021	9.41697

DFFRNQX1

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Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	0
x	1	x	IQ

Footprint

Cell Name	Area
DFFRNQX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
DFFRNQX1	0.01013	0.03180	0.02339	5.48149

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RR)	0.18144	0.82896	7.43531

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RF)	0.22190	0.88822	7.78349
	RN->Q (FF)	0.13704	0.92821	8.61900

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	0.03467	0.05420	0.75322
	setup	CLK (R)	0.13870	0.19306	0.62250

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	-0.01814	-0.07673	-0.69662
	setup	CLK (R)	0.06037	0.13743	1.83078

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	0.03467	0.05420	0.75322
	setup	CLK (R)	RN	0.13870	0.19306	0.62250

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	-0.01814	-0.07673	-0.69662
	setup	CLK (R)	RN	0.06037	0.13743	1.83078

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	recovery	CLK (R)	0.13157	0.21365	7.16805
	removal	CLK (R)	0.03486	0.04036	0.01790

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	recovery	CLK (R)	D	0.13157	0.21365	7.16805
	removal	CLK (R)	D	0.03486	0.04036	0.01790

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	RN ()	(CLK * D)	0.12090	0.66040	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.10110	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.06398	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.06398	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.17287	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.22484	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10357	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51590	34.52320	34.83630

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78470	23.79070	23.99020
	RN	-0.01415	-0.67505	-8.87998
	RN	24.23150	24.25570	24.60620

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	34.32610	34.32950	34.52350
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	22.75710	22.75620	22.75800
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	22.74020	22.74330	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	14.42940	14.43300	14.61840
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	9.85933	9.85942	9.85927

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	16.53660	16.54490	16.76090
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	23.36840	23.36980	23.37000
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	23.35970	23.36090	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	9.99451	10.00360	10.21160
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	10.43050	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	22.01620	22.01440	22.01630
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	22.01000	22.00810	22.00910
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	13.82010	13.82370	14.02080
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	9.22418	9.22482	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	23.85310	23.85400	23.85450
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	23.85370	23.85490	23.85550
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	10.54250	10.55040	10.74610
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	10.91460	10.91270	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	34.32870	34.32730	34.51390
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	23.89430	23.89530	24.08350
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	23.90720	23.90760	24.10180
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	23.89200	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	**(D * RN * Q)**	0.00000	0.00000	0.00000
	(D * RN * Q)	16.55520	16.56010	16.77150
	(D * RN * !Q)	0.00000	0.00000	0.00000
	(D * RN * !Q)	13.80880	13.81470	13.99940
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	9.21176	9.21577	9.40995
	(!D * RN * Q)	0.00000	0.00000	0.00000
	(!D * RN * Q)	13.90110	13.90990	14.13170
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	9.21743	9.22344	9.42132
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	9.21279	9.22021	9.41697

DFFRNX1

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Truth Table

INPUT			OUTPUT	
D	RN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	0	1
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFRNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	RN	CLK	Q	QN
DFFRNX1	0.01013	0.03170	0.02337	5.44593	3.68531

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RR)	0.18150	0.82668	7.40006
	QN->Q (FR)	0.05057	0.83099	10.46690

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RF)	0.23902	1.61692	17.47750
	QN->Q (RF)	0.05541	0.79494	9.94309
	RN->Q (FF)	0.11862	1.70427	21.28830

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RR)	0.17038	0.68348	5.14395
	Q->QN (FR)	0.04751	0.72175	8.39427
	RN->QN (FR)	0.04953	0.73595	8.49577

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RF)	0.27035	1.69369	15.50860
	Q->QN (RF)	0.06804	0.90178	10.11430

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	0.03745	0.05392	0.72606
	setup	CLK (R)	0.14177	0.19399	0.62221

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	-0.01814	-0.07673	-0.69662
	setup	CLK (R)	0.06127	0.13429	1.81319

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	0.03745	0.05392	0.72606
	setup	CLK (R)	RN	0.14177	0.19399	0.62221

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	-0.01814	-0.07673	-0.69662
	setup	CLK (R)	RN	0.06127	0.13429	1.81319

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	recovery	CLK (R)	0.13178	0.21346	7.19354
	removal	CLK (R)	0.03486	0.04036	0.01790

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	recovery	CLK (R)	D	0.13178	0.21346	7.19354
	removal	CLK (R)	D	0.03486	0.04036	0.01790

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	RN ()	(CLK * D)	0.12090	0.66040	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.10110	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.07883	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.07883	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.18029	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.15802	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.22484	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10357	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25820	17.25930	17.35230

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89220	11.89480	11.99830
	RN	-0.00707	-0.33621	-4.41119
	RN	12.11650	12.12300	12.21910

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89200	11.89500	11.99670
	RN	-0.00707	-0.26598	-2.98509
	RN	12.11670	12.12290	12.22650

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25830	17.25960	17.35220

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	34.32610	34.32940	34.52350
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	22.75710	22.75620	22.75800
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	22.74020	22.74330	22.74400
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	18.84320	18.84660	19.03820
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	9.85931	9.85943	9.85928

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	16.53650	16.54490	16.76090
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	23.36840	23.36980	23.37000
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	23.35970	23.36090	23.36120
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	13.88370	13.89270	14.10360
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	10.43050	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	22.01830	22.01430	22.01630
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	22.01000	22.00810	22.00910
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	13.82020	13.82370	14.02080
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	9.22417	9.22483	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	23.85300	23.85400	23.85450
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	23.85360	23.85490	23.85550
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	10.54250	10.55040	10.74640
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	10.91460	10.91270	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	34.32870	34.32730	34.51390
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	23.89430	23.89520	24.08350
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	23.90720	23.90760	24.10180
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	23.89200	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	16.55550	16.56040	16.77180
	(D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * !Q * QN)	13.80880	13.81450	13.99940
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	9.21175	9.21576	9.40994
	(!D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * Q * !QN)	13.90110	13.90990	14.13170
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	9.21745	9.22344	9.42133
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	9.21281	9.22022	9.41698

DFFSNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	0
x	1	x	IQN

Footprint

Cell Name	Area
DFFSNQNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
DFFSNQNX1	0.01015	0.02231	0.02229	5.23349

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RR)	0.15611	0.76038	6.98607

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RF)	0.17319	0.84515	7.47287
	SN->QN (FF)	0.26133	0.97571	8.39700

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	0.01494	0.02418	0.89241
	setup	CLK (R)	0.12418	0.18586	1.87390

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	-0.06952	-0.15136	-1.14884
	setup	CLK (R)	0.10518	0.19199	1.64964

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	0.01494	0.02418	0.89241
	setup	CLK (R)	SN	0.12418	0.18586	1.87390

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	-0.06952	-0.15136	-1.14884
	setup	CLK (R)	SN	0.10518	0.19199	1.64964

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	recovery	CLK (R)	0.03712	0.03636	0.32069
	removal	CLK (R)	-0.01690	-0.01261	-0.12408

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	recovery	CLK (R)	!D	0.03712	0.03636	0.32069
	removal	CLK (R)	!D	-0.01690	-0.01261	-0.12408

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	SN ()	(CLK * D)	0.07388	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.06893	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.10605	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.19514	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.11347	0.66040	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36220	35.36950	35.70850

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64100	19.64650	19.86550
	SN	-0.01415	-0.65655	-8.47822
	SN	19.00010	19.02180	19.37080

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	33.71250	33.71300	33.71360
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	19.80280	19.80620	20.00790
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	19.79890	19.80280	20.00590
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	10.28740	10.29230	10.49350

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	34.35940	34.36020	34.36010
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	9.55497	9.56383	9.77692
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	9.55928	9.56831	9.78784
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.91830	14.92850	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	18.32530	18.33120	18.33270
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	9.86791	9.87159	9.87298
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	9.87714	9.87558	9.87851
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	14.41100	14.41370	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	19.55150	19.55540	19.55620
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	11.02560	11.02730	11.02780
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	11.02790	11.02760	11.02850
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	3.67789	3.68731	3.89217

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	19.79670	19.80090	20.01980
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	19.79220	19.79640	20.01260
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	35.13990	35.14270	35.34240
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	10.63660	10.65050	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * QN)	0.00000	0.00000	0.00000
	(D * SN * QN)	13.00140	13.00820	13.24300
	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	9.56583	9.58221	9.85093
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	9.55802	9.57298	9.81653
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	17.39160	17.39690	17.60760
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	14.92440	14.93410	15.17980
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	3.48809	3.49951	3.80064

DFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	**SN**	**CLK**	**Q**
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
DFFSNQX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	**SN**	**CLK**	**Q**
DFFSNQX1	0.01015	0.02175	0.02229	3.66982

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
DFFSNQX1	0.00000	30.10480	57.62730

Delay Information
Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RR)	0.12625	0.65341	5.03573
	SN->Q (FR)	0.05049	0.74449	8.56161

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RF)	0.23675	0.98662	7.74574

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	0.01164	0.02018	0.78577
	setup	CLK (R)	0.12833	0.20262	3.30055

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	-0.06528	-0.14936	-1.15035
	setup	CLK (R)	0.10185	0.18820	1.61202

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	0.01164	0.02018	0.78577
	setup	CLK (R)	SN	0.12833	0.20262	3.30055

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	-0.06528	-0.14936	-1.15035
	setup	CLK (R)	SN	0.10185	0.18820	1.61202

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	recovery	CLK (R)	0.03437	0.03090	4.63758
	removal	CLK (R)	-0.01690	-0.01261	-0.12508

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	recovery	CLK (R)	!D	0.03437	0.03090	4.63758
	removal	CLK (R)	!D	-0.01690	-0.01261	-0.12508

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.07635	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.07635	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.11100	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13822	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.20256	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10852	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64070	19.65130	19.96210
	SN	18.99790	19.01290	19.26980

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36370	35.36940	35.57670

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN * Q) + (!CLK * !SN * Q)	19.80260	19.80600	20.00770
	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	33.71250	33.71300	33.71360
	(CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !SN * Q)	19.79890	19.80270	20.00590
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.27430	14.27840	14.47990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN * Q) + (!CLK * !SN * Q)	9.55498	9.56382	9.77692
	(CLK * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * SN * !Q)	34.35940	34.36030	34.36010
	(CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !SN * Q)	9.55927	9.56830	9.78784
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	19.42220	19.43360	19.64740

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	18.32510	18.33120	18.33270
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	9.86778	9.87158	9.87298
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	9.87714	9.87557	9.87851
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	14.41100	14.41380	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	19.55140	19.55540	19.55620
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	11.02560	11.02730	11.02780
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	11.02790	11.02750	11.02850
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	3.67725	3.68671	3.89154

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	19.79670	19.80090	20.01980
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	19.79220	19.79640	20.01260
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	35.13990	35.14290	35.34250
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	10.63660	10.65050	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	9.56603	9.58219	9.85092
	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	13.00140	13.00820	13.24300
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	9.55800	9.57297	9.81652
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	14.92440	14.93410	15.17980
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	17.39290	17.39690	17.60760
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	3.48864	3.49951	3.80064

DFFSNRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0
x	1	1	x	IQ

Footprint

Cell Name	Area
DFFSNRNQX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
DFFSNRNQX1	0.01012	0.03417	0.02266	0.02233	3.55003

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RR)	0.17480	0.70010	5.08583
	SN->Q (FR)	0.05059	0.73430	8.45663

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RF)	0.24911	0.97642	7.38302
	RN->Q (FF)	0.13311	0.91735	8.08296
	SN->Q (RF)	0.07379	0.89378	9.84027

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.01291	-0.00816	2.04406
	setup	CLK (R)	0.16770	0.21744	0.69127

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.06116	-0.14127	-1.08690
	setup	CLK (R)	0.10343	0.19073	2.28700

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.01291	-0.00816	2.04406
	setup	CLK (R)	(RN * SN)	0.16770	0.21744	0.69127

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06116	-0.14127	-1.08690
	setup	CLK (R)	(RN * SN)	0.10343	0.19073	2.28700

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.16059	0.24257	7.63089
	removal	CLK (R)	-0.01278	-0.02018	-0.06455
	hold	SN (R)	0.02512	0.06307	0.24942
	setup	SN (R)	0.01482	0.03027	0.96075

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.16059	0.24257	7.63089
	removal	CLK (R)	(D * SN)	-0.01278	-0.02018	-0.06455
	hold	SN (R)	(CLK * D)	0.00157	-0.02018	-0.10605
	hold	SN (R)	(CLK * !D)	0.00157	-0.02018	-0.10755
	hold	SN (R)	(!CLK * D)	0.02512	0.06307	0.22993
	hold	SN (R)	(!CLK * !D)	0.02365	0.06054	0.24942
	setup	SN (R)	(CLK * D)	0.00601	0.02775	0.56078
	setup	SN (R)	(CLK * !D)	0.00773	0.03027	0.67163
	setup	SN (R)	(!CLK * D)	0.01482	0.01096	0.96075
	setup	SN (R)	(!CLK * !D)	0.01309	0.01120	0.80310

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11100	0.66040	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10852	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.07388	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.07388	0.66040	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.02351	0.01181	4.39738
	removal	CLK (R)	0.00262	0.01009	0.01218
	hold	RN (R)	0.04584	0.10848	0.63705
	setup	RN (R)	0.02540	0.01736	0.57537

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(!D * RN)	0.02351	0.01181	4.39738
	removal	CLK (R)	(!D * RN)	0.00262	0.01009	0.01218
	hold	RN (R)	CLK	0.03781	0.10848	0.63705
	hold	RN (R)	(!CLK * D)	0.04580	0.10091	0.57136
	hold	RN (R)	(!CLK * !D)	0.04584	0.09839	0.59271
	setup	RN (R)	CLK	0.02540	0.01736	0.49059
	setup	RN (R)	(!CLK * D)	-0.00092	0.00505	0.57537
	setup	RN (R)	(!CLK * !D)	-0.00457	0.00505	0.16519

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08625	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08625	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.08625	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.08625	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.16792	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23226	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11842	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.67180	32.67860	32.94930
	SN	32.59690	32.60890	32.85660

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.50660	32.51260	32.71460
	RN	-0.01415	-0.52015	-5.75100
	RN	33.06310	33.08420	33.42670
	SN	32.67600	32.67970	32.87860

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	32.63820	32.64090	32.83720
	(CLK * RN * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q)	31.10940	31.11070	31.11060
	(CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q)	32.63690	32.63980	32.83760
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	31.10010	31.10060	31.10100
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.56890	17.57280	17.76560
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	17.19890	17.20060	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)	15.79770	15.80730	16.01610
	(CLK * RN * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q)	31.73570	31.73480	31.73490
	(CLK * RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q)	15.80330	15.80970	16.02450
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)	31.72850	31.72830	31.72880
	(!CLK * RN * SN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN)	17.58330	17.59300	17.80420
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)	17.79770	17.79870	17.79850

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	30.36200	30.36400	30.36250
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	32.88940	32.89110	33.27330
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	30.36060	30.35610	30.35860
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	17.62410	17.62510	17.90980
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	16.91600	16.91930	17.10820
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	14.12260	14.12540	14.46640
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	16.50730	16.50630	16.50620
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	3.61086	3.61355	3.78112

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	32.24040	32.24300	32.24330
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	4.84062	4.86691	5.46298
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	32.24290	32.24430	32.24480
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	6.13207	6.14776	6.54180
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	18.12320	18.13170	18.32370
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	0.10887	0.12999	0.55824
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	18.30350	18.30290	18.30280
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	0.08285	0.09277	0.28323

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	30.73470	30.73980	30.74070
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	16.85920	16.86450	16.86550
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	16.87000	16.86660	16.87170
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	17.58620	17.59140	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	31.98780	31.99680	31.99740
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	18.05780	18.06390	18.06460
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	18.06320	18.06440	18.06470
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	6.06930	6.07912	6.28628

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	32.63360	32.63570	32.82960
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	32.63310	32.63480	32.82630
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	32.44670	32.45060	32.64070
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	3.97135	3.97221	4.11832
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	32.45680	32.45950	32.65370
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	18.13460	18.14630	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	15.81220	15.82550	16.06380
	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	16.61310	16.61580	16.82590
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	15.80670	15.81840	16.03970
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	16.18710	16.19310	16.39260
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	0.08312	0.09133	0.29352
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	17.59960	17.60690	17.85610
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	16.18920	16.19600	16.40700
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	5.63916	5.64933	5.93703

DFFSNRNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT	
D	RN	SN	CLK	Q	QN
0	1	1	R	0	1
1	1	1	R	1	0
x	0	0	x	1	1
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNRNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	D	RN	SN	CLK	Q	QN
DFFSNRNX1	0.01012	0.03429	0.02266	0.02233	3.55003	3.57679

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RR)	0.17484	0.69963	5.08914
	QN->Q (FR)	0.05885	0.72617	8.26726
	SN->Q (FR)	0.05080	0.73648	8.45708

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RF)	0.26544	1.49760	13.33210
	QN->Q (RF)	0.08644	0.82456	8.72930
	RN->Q (FF)	0.14699	1.58471	16.98700
	SN->Q (RF)	0.07379	0.89385	9.84025

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RR)	0.16794	0.67311	5.02463
	Q->QN (FR)	0.04755	0.71330	8.24992
	RN->QN (FR)	0.05031	0.73576	8.47312

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RF)	0.26456	1.65217	15.00370
	Q->QN (RF)	0.06750	0.88166	9.81575
	RN->QN (RF)	0.26914	1.08564	10.29660
	SN->QN (FF)	0.13628	1.64921	17.89340

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.01288	-0.00819	2.02883
	setup	CLK (R)	0.16769	0.21750	0.69402

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.06116	-0.14127	-1.08824
	setup	CLK (R)	0.10265	0.18863	2.19846

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.01288	-0.00819	2.02883
	setup	CLK (R)	(RN * SN)	0.16769	0.21750	0.69402

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.06116	-0.14127	-1.08824
	setup	CLK (R)	(RN * SN)	0.10265	0.18863	2.19846

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.16032	0.24294	7.64192
	removal	CLK (R)	-0.01278	-0.02018	-0.06455
	hold	SN (R)	0.03083	0.07820	0.48988
	setup	SN (R)	0.02869	0.04036	1.33841

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(D * SN)	0.16032	0.24294	7.64192
	removal	CLK (R)	(D * SN)	-0.01278	-0.02018	-0.06455
	hold	SN (R)	(CLK * D)	0.00157	-0.02018	-0.10605
	hold	SN (R)	(CLK * !D)	0.00157	-0.02018	-0.10755
	hold	SN (R)	(!CLK * D)	0.03083	0.07820	0.47943
	hold	SN (R)	(!CLK * !D)	0.02834	0.07820	0.48988
	setup	SN (R)	(CLK * D)	0.00601	0.02775	0.49996
	setup	SN (R)	(CLK * !D)	0.00652	0.03027	0.59465
	setup	SN (R)	(!CLK * D)	0.02717	0.04036	1.33841
	setup	SN (R)	(!CLK * !D)	0.02869	0.04036	1.05500

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11347	0.66040	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.11100	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.08378	0.66040	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.08625	0.66040	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.02474	0.01156	4.13525
	removal	CLK (R)	0.00262	0.01009	0.01218
	hold	RN (R)	0.05787	0.12613	0.68813
	setup	RN (R)	0.02524	0.01742	0.55640

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(!D * RN)	0.02474	0.01156	4.13525
	removal	CLK (R)	(!D * RN)	0.00262	0.01009	0.01218
	hold	RN (R)	CLK	0.04029	0.10848	0.63705
	hold	RN (R)	(!CLK * D)	0.05787	0.12361	0.66457
	hold	RN (R)	(!CLK * !D)	0.05783	0.12613	0.68813
	setup	RN (R)	CLK	0.02524	0.01742	0.48980
	setup	RN (R)	(!CLK * D)	-0.01303	-0.02270	0.55640
	setup	RN (R)	(!CLK * !D)	-0.01451	-0.02523	0.10098

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08625	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08625	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.09615	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.09615	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.17534	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.15802	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23226	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11595	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33600	16.33760	16.43300
	SN	-0.00707	-0.26008	-2.87550
	SN	16.29920	16.30470	16.40190

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25320	16.25570	16.35680
	RN	-0.00707	-0.26008	-2.87550
	RN	16.53140	16.53710	16.64060
	SN	32.67590	32.67970	32.87880

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25320	16.25560	16.35660
	RN	-0.00707	-0.26125	-2.89718
	RN	16.53200	16.53740	16.63980

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33620	16.33790	16.43370
	RN	32.89150	32.89530	33.17100
	SN	-0.00707	-0.26125	-2.89718
	SN	16.29950	16.30370	16.40090

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	32.63820	32.64090	32.83720
	(CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q * QN)	31.10930	31.11060	31.11060
	(CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q * !QN)	32.63720	32.64010	32.83790
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	31.10010	31.10060	31.10090
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	17.57230	17.57630	17.77380
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	17.19890	17.20060	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)	15.79770	15.80730	16.01610
	(CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * SN * !Q * QN)	31.73560	31.73480	31.73490
	(CLK * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !SN * Q * !QN)	15.80330	15.80970	16.02450
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)	31.72870	31.72850	31.72900
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)	17.58800	17.59810	17.81410
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)	17.79770	17.79860	17.79850

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	30.36190	30.36400	30.36250
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	30.36060	30.35610	30.35860
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	16.91610	16.92020	17.10820
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	16.50730	16.50630	16.50610

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	32.24030	32.24300	32.24320
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	32.24290	32.24430	32.24480
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	18.12310	18.13170	18.32360
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	18.30340	18.30290	18.30280

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	30.73470	30.73980	30.74070
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	16.85920	16.86450	16.86550
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	16.87000	16.86660	16.87170
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	17.58620	17.59140	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	31.98780	31.99680	31.99740
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	18.05780	18.06390	18.06460
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	18.06330	18.06440	18.06470
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	6.06910	6.07893	6.28575

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	32.63360	32.63590	32.82970
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	32.63310	32.63480	32.82630
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	32.44660	32.45050	32.64070
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	3.97133	3.97219	4.11830
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	32.45680	32.45940	32.65360
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	18.13460	18.14630	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	15.81220	15.82550	16.06380
	(D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q * QN)	16.61280	16.61550	16.82550
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	15.80670	15.81840	16.03970
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	16.18710	16.19310	16.39260
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	0.08310	0.09131	0.29350
	(!D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q * !QN)	17.59960	17.60690	17.85600
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	16.18890	16.19570	16.40660
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	5.63913	5.64932	5.93703

DFFSNX1

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Truth Table

INPUT			OUTPUT	
D	SN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	1	0
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	SN	CLK	Q	QN
DFFSNX1	0.01015	0.02175	0.02229	3.67547	5.23618

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RR)	0.12601	0.65290	5.04865
	QN->Q (FR)	0.05829	0.73537	8.43143
	SN->Q (FR)	0.05025	0.74362	8.60439

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RF)	0.25309	1.52349	13.94630
	QN->Q (RF)	0.08562	0.83974	9.00544

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RR)	0.15623	0.76005	7.00273
	Q->QN (FR)	0.04428	0.80864	10.18930

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RF)	0.19106	1.65629	18.10340
	Q->QN (RF)	0.04593	0.80024	10.03750
	SN->QN (FF)	0.11458	1.72150	21.30640

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	0.01164	0.02018	0.74976
	setup	CLK (R)	0.12822	0.20325	3.30595

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	-0.06522	-0.14936	-1.15032
	setup	CLK (R)	0.09767	0.18636	1.59617

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	0.01164	0.02018	0.74976
	setup	CLK (R)	SN	0.12822	0.20325	3.30595

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	-0.06522	-0.14936	-1.15032
	setup	CLK (R)	SN	0.09767	0.18636	1.59617

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	recovery	CLK (R)	0.03170	0.02995	4.32719
	removal	CLK (R)	-0.01690	-0.01261	-0.12508

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	recovery	CLK (R)	!D	0.03170	0.02995	4.32719
	removal	CLK (R)	!D	-0.01690	-0.01261	-0.12508

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	SN ()	(CLK * D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07140	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.08378	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.08378	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.11595	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15060	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.20256	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10605	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.81951	9.82319	9.93527
	SN	-0.00707	-0.26555	-2.97713
	SN	9.50011	9.50524	9.60658

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68170	17.68380	17.78800

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68140	17.68380	17.78760

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.82043	9.82369	9.93668
	SN	-0.00707	-0.32838	-4.24129
	SN	9.49978	9.50498	9.59971

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	19.80260	19.80600	20.00770
	(CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * SN * !Q * QN)	33.71240	33.71300	33.71360
	(CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * Q * !QN)	19.79890	19.80270	20.00590
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	10.28690	10.29190	10.49300

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)	9.55495	9.56380	9.77689
	(CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * SN * !Q * QN)	34.35940	34.36020	34.36010
	(CLK * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * Q * !QN)	9.55924	9.56827	9.78781
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)	14.91830	14.92840	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	18.32430	18.33110	18.33250
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	9.86764	9.87164	9.87330
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	9.87687	9.87543	9.87827
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	14.41000	14.41320	14.60650

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	19.55130	19.55540	19.55620
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	11.02530	11.02720	11.02770
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	11.02780	11.02750	11.02780
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	3.67757	3.68698	3.89183

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	19.79660	19.80080	20.01960
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	19.79220	19.79640	20.01260
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	35.14010	35.14310	35.34260
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	10.63670	10.65060	10.91050

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	9.56593	9.58228	9.85099
	(D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * SN * !Q * QN)	13.00300	13.00970	13.24410
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	9.55826	9.57304	9.81662
	(!D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * SN * Q * !QN)	14.92450	14.93420	15.17990
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	17.39170	17.39700	17.60770
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	3.48831	3.49973	3.80088

DFFX1

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Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	R	0	1
1	R	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
DFFX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
DFFX1	0.01055	0.02310	5.42033	5.42584

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RR)	0.13085	0.77166	7.19024
	QN->Q (FR)	0.05057	0.83100	10.45400

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RF)	0.22675	1.61397	17.57770
	QN->Q (RF)	0.05570	0.79493	9.93421

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RR)	0.15836	0.77193	7.20227
	Q->QN (FR)	0.04466	0.82105	10.43860

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RF)	0.19622	1.70774	18.81930
	Q->QN (RF)	0.04703	0.82208	10.40220

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	0.04506	0.07673	0.67467
	setup	CLK (R)	0.10810	0.17277	0.78704

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	-0.03300	-0.08682	-0.73801
	setup	CLK (R)	0.05620	0.12309	1.49870

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.12090	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.15060	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.19019	0.66040	16.50020
	min_pulse_width	CLK ()	!D	0.09367	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56510	10.56760	10.67500

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12880	13.13140	13.23750

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12810	13.13130	13.23580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56540	10.56790	10.67660

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	21.09440	21.10050	21.31620
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	24.93600	24.93560	24.93790
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.29550	11.30240	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	10.10440	10.11320	10.34910
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	25.56960	25.56910	25.56920
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.30960	11.31810	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	21.09580	21.09810	21.30850
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	26.21560	26.21810	26.41820

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	**mid**	**last**
DFFX1	**(D * Q * !QN)**	0.00000	0.00000	0.00000
	(D * Q * !QN)	10.11640	10.12240	10.35220
	(D * !Q * QN)	0.00000	0.00000	0.00000
	(D * !Q * QN)	10.32690	10.33310	10.54020
	(!D * Q * !QN)	0.00000	0.00000	0.00000
	(!D * Q * !QN)	11.32140	11.33020	11.55020
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	9.90683	9.91142	10.11400

DLATCHN

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	**GATE_N**	**Q**
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
DLATCHN	179.87680

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	**GATE_N**	**Q**
DLATCHN	0.02276	0.01036	2.88873

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
DLATCHN	0.00000	22.21240	28.71060

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (RR)	0.18724	0.93039	6.98398
	GATE_N->Q (-R)	0.23686	1.05826	8.01314

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (FF)	0.15972	0.53260	3.42270
	GATE_N->Q (-F)	0.17486	0.57608	3.50583

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	13.54570	13.56870	14.09900
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	13.29360	13.30530	13.53960

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	9.19406	9.22145	9.74499
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	7.78960	7.79889	8.08816

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	16.22050	16.23230	16.47220

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	11.66530	11.67830	11.91190

Passive power(pJ) for GATE_N rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	16.31890	16.32900	16.59530
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	11.95670	11.96630	12.20380

Passive power(pJ) for GATE_N falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	11.77200	11.78740	12.03900
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.98989	9.00602	9.25037

DLATCH

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	GATE	Q
x	0	1
0	1	0
1	1	1

Footprint

Cell Name	Area
DLATCH	162.64960

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE	Q
DLATCH	0.02279	0.02079	2.90667

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCH	0.00000	18.14490	24.27320

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (RR)	0.18665	0.92887	6.99745
	GATE->Q (-R)	0.18598	0.91495	6.90747

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (FF)	0.15826	0.53179	3.43223
	GATE->Q (-F)	0.11870	0.44685	2.63781

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	14.12970	14.15360	14.69420
	GATE	0.00000	0.00000	0.00000
	GATE	13.88780	13.89630	14.14810

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	9.55085	9.58113	10.10720
	GATE	0.00000	0.00000	0.00000
	GATE	8.11930	8.12739	8.41170

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	12.15630	12.16730	12.40740

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	8.01455	8.02758	8.26039

Passive power(pJ) for GATE rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.37340	12.38320	12.61050
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.32390	9.33338	9.55308

Passive power(pJ) for GATE falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.18760	12.20220	12.45500
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.23660	8.24815	8.47690

FA

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT	
A	B	CIN	COUT	SUM
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
FA	309.08081

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	A	B	CIN	COUT	SUM
FA	0.03793	0.03735	0.03457	5.74192	2.61746

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
FA	0.00000	36.98230	51.95610

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (RR)	0.32464	0.88981	7.02011
	B->COUT (RR)	0.34861	0.91691	7.07676
	CIN->COUT (RR)	0.14765	0.72344	6.91082

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (FF)	0.29870	0.81146	6.00401
	B->COUT (FF)	0.30839	0.82434	6.09275
	CIN->COUT (FF)	0.17121	0.68898	5.84574

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-R)	-	0.25844	1.10947	8.34626
	B->SUM (-R)	-	0.24113	1.08475	8.11268
	CIN->SUM (RR)	(A * B) + (!A * !B)	0.10813	0.81105	6.60626
	CIN->SUM (FR)	(A * !B) + (!A * B)	0.07131	0.96902	9.99325

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-F)	-	0.23323	0.80278	5.31322
	B->SUM (-F)	-	0.21528	0.78016	5.14248
	CIN->SUM (FF)	(A * B) + (!A * !B)	0.08493	0.57865	4.52819
	CIN->SUM (RF)	(A * !B) + (!A * B)	0.04526	0.66078	7.00185

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	17.40010	17.41560	17.77020
	B	17.39970	17.41490	17.77210
	CIN	0.00000	0.00000	0.00000
	CIN	14.35090	14.36130	14.61390

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	5.75188	5.75882	5.87941
	B	5.75555	5.76095	5.87764
	CIN	0.00000	0.00000	0.00000
	CIN	3.75753	3.77022	4.02641

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	26.85160	26.88730	27.56340
	B	-	0.00000	0.00000	0.00000
	B	-	26.85530	26.88750	27.55930
	CIN	(A * B) + (!A * !B)	0.00000	0.00000	0.00000
	CIN	(A * B) + (!A * !B)	27.04000	27.04910	27.28340
	CIN	(A * !B) + (!A * B)	0.00000	0.00000	0.00000
	CIN	(A * !B) + (!A * B)	3.75693	3.77111	4.03522

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	28.07800	28.28470	29.73710
	B	-	0.00000	0.00000	0.00000
	B	-	28.07530	28.23050	29.23270
	CIN	(A * B) + (!A * !B)	0.00000	0.00000	0.00000
	CIN	(A * B) + (!A * !B)	33.67420	33.68630	33.91570
	CIN	(A * !B) + (!A * B)	0.00000	0.00000	0.00000
	CIN	(A * !B) + (!A * B)	14.35090	14.36140	14.62400

HA

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT	
A	B	COUT	SUM
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Footprint

Cell Name	Area
HA	136.80881

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	A	B	COUT	SUM
HA	0.03356	0.03772	6.17943	2.60808

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
HA	0.00000	17.03700	32.46250

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (RR)	0.07164	0.66765	7.65971
	B->COUT (RR)	0.06558	0.67175	7.70589

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (FF)	0.06413	0.58942	6.29787
	B->COUT (FF)	0.06058	0.57840	6.25763

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (RR)	!B	0.08570	0.78315	6.48830
	A->SUM (FR)	B	0.07751	0.98728	10.08150
	B->SUM (RR)	!A	0.10801	0.79363	6.42662
	B->SUM (FR)	A	0.06954	0.97696	10.05620

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (FF)	!B	0.07743	0.56543	4.61616
	A->SUM (RF)	B	0.04680	0.63124	6.57545
	B->SUM (FF)	!A	0.08584	0.58723	4.58569
	B->SUM (RF)	A	0.04131	0.65159	6.90808

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	11.85120	11.86460	12.13420
	B	0.00000	0.00000	0.00000
	B	11.85340	11.86470	12.14100

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	0.86447	0.88089	1.14894
	B	0.00000	0.00000	0.00000
	B	0.85658	0.87123	1.11641

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	0.86466	0.88134	1.16340
	A	!B	0.00000	0.00000	0.00000
	A	!B	3.99926	4.01192	4.26147
	B	A	0.00000	0.00000	0.00000
	B	A	0.85679	0.87142	1.12578
	B	!A	0.00000	0.00000	0.00000
	B	!A	4.01497	4.02317	4.28242

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	11.85070	11.86580	12.15270
	A	!B	0.00000	0.00000	0.00000
	A	!B	9.43961	9.45346	9.69810
	B	A	0.00000	0.00000	0.00000
	B	A	11.85090	11.86560	12.16250
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.44589	9.45976	9.67637

INVX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
INVX1	24.83200

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
INVX1	0.01040	5.57495

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
INVX1	0.00000	3.90439	7.80687

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (FR)	0.02647	0.76036	9.91760

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (RF)	0.02008	0.56880	7.36203

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	0.01840	0.02343	0.08289

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	4.60766	4.60859	4.63402

MUX2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A0	A1	S	Y
0	0	x	0
x	1	0	1
0	1	1	0
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
MUX2X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A0	A1	S	Y
MUX2X1	0.01056	0.01054	0.02177	5.42552

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
MUX2X1	0.00000	13.86620	17.36010

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (RR)	-	0.07915	0.65289	6.88313
	A1->Y (RR)	-	0.07223	0.64656	6.84105
	S->Y (RR)	(A0 * !A1)	0.08305	0.63928	6.76020
	S->Y (FR)	(!A0 * A1)	0.11518	0.72241	7.19830

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (FF)	-	0.07857	0.77176	8.29389
	A1->Y (FF)	-	0.07238	0.74663	8.18526
	S->Y (FF)	(A0 * !A1)	0.08111	0.77138	8.25537
	S->Y (RF)	(!A0 * A1)	0.10376	0.68597	6.75608

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.83644	9.84644	10.09650
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.46015	5.47269	5.71171
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	10.13200	10.15660	10.69710
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	5.71799	5.73259	5.96212

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.85208	9.86542	10.08950
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.47650	5.49039	5.72997
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	5.14857	5.18042	5.74431
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	9.55731	9.56951	9.80113

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A1 * !S * Y) + (!A1 * !S * !Y)	0.00000	0.00000	0.00000
	(A1 * !S * Y) + (!A1 * !S * !Y)	5.12255	5.12211	5.12366

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A1 * !S * Y) + (!A1 * !S * !Y)	0.00000	0.00000	0.00000
	(A1 * !S * Y) + (!A1 * !S * !Y)	5.70502	5.70543	5.70573

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * S * Y) + (!A0 * S * !Y)	0.00000	0.00000	0.00000
	(A0 * S * Y) + (!A0 * S * !Y)	9.47496	9.47261	9.47443

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * S * Y) + (!A0 * S * !Y)	0.00000	0.00000	0.00000
	(A0 * S * Y) + (!A0 * S * !Y)	10.08630	10.08610	10.08610

Passive power(pJ) for S rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * A1 * Y)	0.00000	0.00000	0.00000
	(A0 * A1 * Y)	9.89291	9.91297	10.41400
	(!A0 * !A1 * !Y)	0.00000	0.00000	0.00000
	(!A0 * !A1 * !Y)	9.84636	9.85680	10.09490

Passive power(pJ) for S falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * A1 * Y)	0.00000	0.00000	0.00000
	(A0 * A1 * Y)	5.47503	5.50517	5.99341
	(!A0 * !A1 * !Y)	0.00000	0.00000	0.00000
	(!A0 * !A1 * !Y)	5.42791	5.44268	5.68488

NAND2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
NAND2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NAND2X1	0.01054	0.01061	5.10239

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND2X1	0.00000	3.89237	15.56580

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (FR)	0.03423	0.77443	9.79964
	B->Y (FR)	0.03011	0.76659	9.73368

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (RF)	0.03766	0.77009	9.66751
	B->Y (RF)	0.03028	0.74464	9.37707

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	0.02835	0.03200	0.08023
	B	0.00000	0.00000	0.00000
	B	0.01897	0.02257	0.05611

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	9.73372	9.73413	9.74473
	B	0.00000	0.00000	0.00000
	B	9.73374	9.73470	9.74941

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	-0.00180	-0.00186	-0.00180

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01092	0.01054	0.01036

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.00105	0.00100	0.00101

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.01104	0.01118	0.01126

NAND3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
NAND3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NAND3X1	0.01059	0.01035	0.01056	3.37535

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND3X1	0.00000	2.91632	23.32440

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (FR)	0.04043	0.69041	7.93026
	B->Y (FR)	0.03749	0.68263	7.88655
	C->Y (FR)	0.03270	0.67795	7.88261

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (RF)	0.06018	0.83998	9.05898
	B->Y (RF)	0.05277	0.82366	9.05906
	C->Y (RF)	0.04324	0.81268	9.14826

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	0.03782	0.04129	0.09703
	B	0.00000	0.00000	0.00000
	B	0.02909	0.03231	0.08476
	C	0.00000	0.00000	0.00000
	C	0.02036	0.02368	0.07328

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	14.71990	14.72060	14.73040
	B	0.00000	0.00000	0.00000
	B	14.72010	14.72050	14.73060
	C	0.00000	0.00000	0.00000
	C	14.72060	14.72110	14.73550

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	-0.00242	-0.00246	-0.00243
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.00067	0.00061	0.00062
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	-0.00451	-0.00453	-0.00456

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.01128	0.01066	0.01061
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.01273	0.01231	0.01225
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	0.00996	0.00996	0.00995

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	-0.00212	-0.00215	-0.00219
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.00425	0.00423	0.00422
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	-0.00467	-0.00470	-0.00473

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.01082	0.01031	0.01029
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.01201	0.01215	0.01215
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	0.01103	0.01116	0.01116

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.00019	0.00012	0.00010
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.00448	0.00442	0.00444

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01100	0.01120	0.01122
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.01079	0.01095	0.01089

NOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
x	1	0
1	x	0

Footprint

Cell Name	Area
NOR2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NOR2X1	0.01081	0.01051	2.54971

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR2X1	0.00000	2.00110	6.04921

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (FR)	0.06116	0.94326	9.66809
	B->Y (FR)	0.04489	0.85097	9.03151

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (RF)	0.02593	0.47060	4.90216
	B->Y (RF)	0.02296	0.46390	4.87768

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	0.03273	0.03364	0.06680
	B	0.00000	0.00000	0.00000
	B	0.02729	0.03123	0.09585

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	2.45448	2.45594	2.48863
	B	0.00000	0.00000	0.00000
	B	2.72034	2.71783	2.74573

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.06819	0.06685	0.06652

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	3.67762	3.67728	3.67695

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	0.01664	0.01663	0.01663

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	2.45799	2.45770	2.45805

OR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
OR2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
OR2X1	0.01076	0.01066	5.70062

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR2X1	0.00000	3.08382	5.40294

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (RR)	0.05608	0.57586	6.45627
	B->Y (RR)	0.05318	0.60421	6.74836

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (FF)	0.09469	0.59629	6.23996
	B->Y (FF)	0.07960	0.58613	6.23489

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.85508	1.87304	2.18076
	B	0.00000	0.00000	0.00000
	B	2.93916	2.95162	3.20602

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.25422	1.26143	1.47467
	B	0.00000	0.00000	0.00000
	B	2.95782	2.97255	3.23968

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.04885	0.04743	0.04726

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	3.12398	3.12372	3.12350

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	0.01974	0.01973	0.01974

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	1.84553	1.84547	1.84610

OR3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	0
0	x	1	1
x	1	x	1
1	x	x	1

Footprint

Cell Name	Area
OR3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
OR3X1	0.01077	0.01055	0.01054	5.59917

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR3X1	0.00000	1.61262	5.01932

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (RR)	0.05988	0.57627	6.24282
	B->Y (RR)	0.05945	0.59605	6.24269
	C->Y (RR)	0.05539	0.61029	6.57160

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (FF)	0.14574	0.65314	6.33036
	B->Y (FF)	0.13265	0.65916	6.37101
	C->Y (FF)	0.10805	0.64252	6.36220

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	1.89624	1.91217	2.20987
	B	0.00000	0.00000	0.00000
	B	1.61743	1.62766	1.84353
	C	0.00000	0.00000	0.00000
	C	2.55368	2.56305	2.75602

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	0.62656	0.63055	0.80224
	B	0.00000	0.00000	0.00000
	B	1.31304	1.31832	1.48721
	C	0.00000	0.00000	0.00000
	C	2.58833	2.59926	2.81567

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01817	0.01744	0.01736
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.06576	0.06442	0.06428
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.04925	0.04755	0.04731

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06099	0.06105	0.06113
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1.53522	1.53526	1.53528
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	2.67785	2.67733	2.67786

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02082	0.02084	0.02087
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.03902	0.03901	0.03900
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.03896	0.03729	0.03698

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06468	0.06470	0.06470
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	1.88000	1.87987	1.88012
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	2.75180	2.75081	2.75146

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.01258	0.01257	0.01256
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	0.01730	0.01729	0.01728

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.08815	0.08816	0.08819
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	1.88339	1.88336	1.88369

TIEHI

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Footprint

Cell Name	Area
TIEHI	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
TIEHI	12.24737

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TIEHI	0.00000	0.00000	0.00000

TIELO

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Footprint

Cell Name	Area
TIELO	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	YN
TIELO	26.69489

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TIELO	0.00000	0.00000	0.00000

TMRDFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
x	x	IQN
1	1	IQN

Footprint

Cell Name	Area
TMRDFFQNX1	584.71600

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
TMRDFFQNX1	0.04012	0.07454	2.70083

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQNX1	0.00000	71.39310	102.09600

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	**CLK->QN (RR)**	0.37945	1.46379	10.48420

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	0.04354	0.06989	0.58191
	setup	CLK (R)	-0.03405	-0.05550	-0.22570

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	-0.02415	-0.05547	-0.49159
	setup	CLK (R)	0.04982	0.10634	1.01811

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	!D	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	!D	0.09367	0.66040	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	76.32920	76.40300	77.72320

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	72660.70000	72662.40000	72662.90000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	72248.70000	72250.90000	72261.10000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71982.40000	71982.80000	71983.80000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	64003.90000	64003.50000	64004.60000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	64576.40000	64579.60000	64581.10000
	!CLK	0.00000	0.00000	0.00000
	!CLK	89969.20000	89970.10000	89980.70000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	63371.30000	63390.10000	63406.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	65506.90000	66195.80000	66661.60000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	76.35860	76.42340	77.69390

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	103226.00000	104138.00000	100651.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	98426.30000	98504.00000	98725.70000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	98885.70000	98870.10000	98889.20000
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	88902.90000	88915.40000	89077.70000

TMRDFFQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
D	**CLK**	**Q**
x	x	-

Footprint

Cell Name	Area
TMRDFFQX1	601.94318

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	**CLK**	**Q**
TMRDFFQX1	0.04023	0.07443	8.45426

Leakage Information

Cell Name	Leakage(nW)		
	Min.	**Avg**	**Max.**
TMRDFFQX1	0.00000	77.76220	108.74900

Delay Information
Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	CLK->Q (RF)	0.44989	1.08458	8.38492

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	0.04308	0.06572	0.55562
	setup	CLK (R)	-0.03405	-0.05550	-0.22520

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	-0.02415	-0.05547	-0.49859
	setup	CLK (R)	0.05238	0.10491	1.01589

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	!D	0.14564	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	!D	0.09120	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	-0.01415	-0.87547	-13.69590
	CLK	80.66470	80.73410	82.01520

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	72259.50000	72261.70000	72269.70000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	72293.80000	72294.00000	72293.00000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71607.90000	71607.50000	71607.40000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	64325.10000	64326.60000	64328.60000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	63863.20000	63863.30000	63864.10000
	!CLK	0.00000	0.00000	0.00000
	!CLK	87402.50000	87396.40000	87405.30000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	65805.60000	66462.90000	66895.50000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	63399.50000	63417.80000	63431.70000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	80.63420	80.69870	81.96450

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	98589.40000	98715.80000	98911.80000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	103371.00000	104246.00000	100554.00000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	88941.00000	88955.60000	89131.60000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	99582.40000	99577.90000	99582.20000

TMRDFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNQX1	670.85199

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
TMRDFFSNQX1	0.04118	0.07157	0.07187	5.32224

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQX1	0.00000	85.26810	134.20100

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RR)	0.25027	0.89428	7.10728
	SN->Q (FR)	0.16641	0.89364	8.02662

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RF)	0.46664	0.97912	5.59773

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	0.00921	0.01618	0.49933
	setup	CLK (R)	0.11989	0.16963	0.98951

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	-0.07014	-0.15388	-1.15314
	setup	CLK (R)	0.09389	0.17763	1.48814

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	0.00921	0.01618	0.49933
	setup	CLK (R)	SN	0.11989	0.16963	0.98951

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	-0.07014	-0.15388	-1.15314
	setup	CLK (R)	SN	0.09389	0.17763	1.48814

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	0.02753	0.02422	3.82793
	removal	CLK (R)	-0.01690	-0.01261	-0.11908

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	!D	0.02753	0.02422	3.82793
	removal	CLK (R)	!D	-0.01690	-0.01261	-0.11908

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.09862	0.66040	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.09615	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.10357	0.66040	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.10110	0.66040	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.14070	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15307	0.66040	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.19514	0.66040	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10357	0.66040	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	59.18060	59.19290	59.84400
	SN	58.46590	58.47710	58.78100

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	105.56500	105.57800	106.21500

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)	100.05900	100.06300	100.06700
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	40.16600	40.17580	40.77120

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)	101.96200	101.96700	101.97200
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	54.53540	54.56320	55.20190

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * Q) + (!CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * D * Q)	55.19030	55.19710	55.19960
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	44.14860	44.15770	44.73060

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(CLK * Q) + (!CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * Q) + (!CLK * D * Q)	58.53240	58.53760	58.54070
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	12.03770	12.06660	12.67380

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(D * SN) + (D * !SN * Q)	0.00000	0.00000	0.00000
	(D * SN) + (D * !SN * Q)	59.21670	59.23050	59.86870
	(!D * SN)	0.00000	0.00000	0.00000
	(!D * SN)	105.39600	105.40500	106.03400
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	33.01070	33.05230	33.83200

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	(D * SN) + (D * !SN * Q)	0.00000	0.00000	0.00000
	(D * SN) + (D * !SN * Q)	40.88170	40.89150	41.58610
	(!D * SN)	0.00000	0.00000	0.00000
	(!D * SN)	53.50100	53.51370	54.13780
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	11.54480	11.57440	12.43390

VOTER3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	x	0
x	1	x	1
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
VOTER3X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
VOTER3X1	0.02221	0.02025	0.02076	5.47682

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTER3X1	0.00000	4.21851	10.36970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (RR)	0.08361	0.68090	6.75559
	B->Y (RR)	0.09738	0.64959	6.03338
	C->Y (RR)	0.09111	0.67400	6.71288

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (FF)	0.14269	0.69252	6.21959
	B->Y (FF)	0.21163	0.83067	6.72540
	C->Y (FF)	0.17707	0.72488	6.41594

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	5.10334	5.11013	5.28426
	B	0.00000	0.00000	0.00000
	B	0.17442	0.17548	0.28527
	C	0.00000	0.00000	0.00000
	C	5.10091	5.10761	5.27322

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	-0.01415	-0.67470	-8.87245
	A	2.09186	2.09867	2.29687
	B	0.00000	0.00000	0.00000
	B	1.12774	1.12702	1.21127
	C	0.00000	0.00000	0.00000
	C	2.10305	2.10875	2.32402

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01491	0.01467	0.01468
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.11450	0.11271	0.11213
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	1.45995	1.45940	1.45937

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	1470.13000	1406.04000	1396.55000
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1863.31000	1843.00000	1835.99000
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22144	2.22144	2.22154

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02720	0.02421	0.02347
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	5.29840	5.38404	5.48488
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	3.15134	3.20755	3.28205

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	5.84790	5.84787	5.84804
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22305	2.22318	2.22334
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.21594	2.21516	2.21547

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.01749	0.01745	0.01752
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	6244.80000	6246.64000	6254.76000
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	6.25238	6.25384	6.25457
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	1.45629	1.45604	1.45520

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.16840	0.16845	0.16849
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	1777.51000	1775.59000	1773.81000
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	9.93310	9.05812	8.84054
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	2.22035	2.21913	2.21946

XNOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Footprint

Cell Name	Area
XNOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XNOR2X1	0.02194	0.02438	2.54427

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XNOR2X1	0.00000	9.81263	18.16860

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (RR)	B	0.08471	0.76294	6.28565
	A->Y (FR)	!B	0.07452	0.98173	9.95692
	B->Y (RR)	A	0.10329	0.82311	6.56840
	B->Y (FR)	!A	0.06300	0.94992	9.80969

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (FF)	B	0.07683	0.55270	4.50661
	A->Y (RF)	!B	0.04974	0.61383	6.30909
	B->Y (FF)	A	0.08910	0.56784	4.51493
	B->Y (RF)	!A	0.04482	0.60473	6.29047

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	5.00170	5.01507	5.27952
	A	!B	0.00000	0.00000	0.00000
	A	!B	0.07237	0.08981	0.37649
	B	A	0.00000	0.00000	0.00000
	B	A	1.83467	1.84363	2.05941
	B	!A	0.00000	0.00000	0.00000
	B	!A	0.06916	0.08833	0.40668

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	11.02970	11.04570	11.30350
	A	!B	0.00000	0.00000	0.00000
	A	!B	8.94410	8.95508	9.18841
	B	A	0.00000	0.00000	0.00000
	B	A	8.99013	9.00379	9.26291
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.34955	9.35916	9.58209

XOR2X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Footprint

Cell Name	Area
XOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XOR2X1	0.02211	0.02441	2.57984

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XOR2X1	0.00000	9.77674	19.21570

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (RR)	!B	0.08493	0.76726	6.33970
	A->Y (FR)	B	0.07678	0.98444	10.00830
	B->Y (RR)	!A	0.10610	0.78356	6.34887
	B->Y (FR)	A	0.06795	0.97252	9.98765

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (FF)	!B	0.07656	0.55941	4.61292
	A->Y (RF)	B	0.04598	0.61843	6.39500
	B->Y (FF)	!A	0.08463	0.58521	4.59993
	B->Y (RF)	A	0.04032	0.64498	6.81881

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	0.54017	0.55561	0.81325
	A	!B	0.00000	0.00000	0.00000
	A	!B	1.73146	1.74545	2.01206
	B	A	0.00000	0.00000	0.00000
	B	A	0.53091	0.54505	0.79763
	B	!A	0.00000	0.00000	0.00000
	B	!A	1.74538	1.75466	2.02172

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	15.38110	15.39620	15.68920
	A	!B	0.00000	0.00000	0.00000
	A	!B	6.22775	6.24253	6.49538
	B	A	0.00000	0.00000	0.00000
	B	A	15.38070	15.39370	15.68900
	B	!A	0.00000	0.00000	0.00000
	B	!A	6.23319	6.24714	6.46674